
EE/CprE/SE 491 WEEKLY REPORT 2

9/22/2026 - 9/29/2026

Group number:

sdmay27-18

Project title: Eddy Current (EC) Probes for Conductivity Measurements

Client &/Advisor:

Advisor – Jiming Song

Client - Dr. Mingyang Lu, CNDE and ISU-ECpE

Team Members/Role:

Connor Crimmins - Client and advisor communication.

Jonathan Gatewood - Software and data-processing coordination.

Eric Williams - Circuit and measurement electronics coordination.

Truong Luu - Team organization, scheduling, and task tracking.

Sebastian De La Torr- Proband sensing hardware coordination.

Benjamin Ponce-Martinez - Testing, calibration, and documentation coordination.

This week our goals were to get going on researching more about our project as well as contacting our advisor and client. For the meeting with Dr. Song this will occur tomorrow at 4:30pm so we can introduce each other as well as discuss a little more about the necessities of this project. We will also find out what Professor Song and Professor Mingyang Lu want to see out of it. For our research the team broke out and began looking more into the project such as what VNAs are and how will we build the system. If we make this an embedded project what will we use as the measure chip and what will hold the memory? These are examples of what we looked into this week to get a better idea of the project. The tasks we completed were just that. For example, we found out that there are certain chips we can use to measure such as the AD5933 which measures impedance and can be converted to admittance to measure conductance. This was a thought or that we can use a Arduino to do the same thing since it already has a memory device attached to it. So far there has not been any changes to our project since we are looking into the project and just trying to understand our project more before we start buying materials.

- Connor Crimmins: worked on looking into what VNA is, what chips to use, and communication with professors.
- Truong Luu: researched the basics of how an Eddy Current probe measures conductivity. He learned how the coil interacts with a metal sample and how a VNA measures its response. This helped him better understand how the project works.
- Jonathan Gatewood: defined what measurements the project needs and established the software/data-processing requirements. The goal is to decide what info should be collected before testing begins so that later analysis, calibration, and validation can be

performed consistently. This plan is intended to connect the physical probe testing with my software and data-analysis responsibilities.

- Eric Williams: researched further into possible coil configurations for the test probes and its connection to the metal to be tested. Some coils researched include the air-cooled pancake coil and the PCB coil. He also looked into the use of a coaxial cable to the AC source that will power the coil. He assisted Sebastian in this research as well.
- Sebastian De La Torre: researched probe coil configurations and sensor fronted requirements for eddy current testing. Eric assisted.
- Benjamin Ponce-Martinez: researched how eddy-current probes can be tested using conductive materials. He also looked into factors that could affect the accuracy of measurements such as probe design or material properties.

<u>NAME</u>	<u>Individual Contributions</u> <i>(Quick list of contributions. This should be short.)</i>	<u>Hours this week</u>	<u>HOURS cumulative</u>
Connor Crimmins	VNA, Chips, Professor Communication	6	8
Jonathan Gatewood	VNA, Data, Processing	6	8
Eric Williams	Coil designs and power connections	6	8
Truong Luu	Research the basic operation of the Eddy Current probe and VNA.	6	8
Sebastian De La Torre	Coil options, sensor requirements.	6	8
Benjamin Ponce-Martinez	Researched how eddy-current probe works, Factors that affect the accuracy of measurements	6	8.5

- Connor Crimmins: will work on finding components that will actually contribute to the project. Discuss further with professors to gain knowledge from them on what to do with the project.
- Truong Luu: will prepare questions about the measurement process for the professor and the client, then continue researching based on their guidance.
- Jonathan Gatewood: will develop the calculation and modeling framework that will support later VNA measurements and conductivity calibration. The main measurements include skin depth, complex impedance, frequency response, and the expected change in the probe signal when the coil is placed near conductive materials. He will also provide equations and example calculations to later be implemented in the data-processing software.
- Eric Williams: will look further into the operation of the AD5933 chip and how it can be applied as a power source and data collection instrument. Connor will assist in this work when possible as well.
- Sebastian De La Torre: will test candidate probe coils in the lab using standard bench equipment to compare basic electrical characteristics and establish baseline readings.
- Benjamin Ponce-Martinez: will determine how testing procedures and results should be documented and organized. This will help when determining our testing and calibration process once we start working with the physical system.